

Case 1

02582 Computational Data Analysis

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Introduction/Data description

The task is to predict a continuous response y from a 100-dimensional feature matrix X given only $n = 100$ labeled observations — a canonical high-dimensional regime ($n = p$) in which standard OLS is undefined and regularization is mandatory. The Expected Prediction Error decomposes as $\text{EPE} = \sigma_\epsilon^2 + \text{Bias}^2 + \text{Var}$, and navigating this bias-variance tradeoff is the central challenge.

The training file `case1Data.csv` contains $n = 100$ rows and 101 columns: one response y and $p = 100$ predictors. Of these, 95 are continuous (`x_01–x_95`) and 5 are ordinal categorical variables (`C_01–C_05`) taking integer values in $\{71, 72, 73, 74, 75\}$. The response y ranges from 39.7 to 375.8 with mean 211.5 and standard deviation 72.5. A correlation analysis reveals a sparse signal: the leading predictor `x_32` achieves $|r| = 0.77$ with y , followed by `x_36` ($|r| = 0.48$) and `x_31` ($|r| = 0.43$), while most other features show weak associations — suggesting that only a handful of predictors carry genuine signal. The training set contains 1,489 missing entries ($\approx 15\%$ per column), and the prediction set `case1Data_Xnew.csv` ($n_{\text{new}} = 1,000$ observations) contains 15,061 missing entries, making imputation a critical design choice.

Model and method

All preprocessing and modelling steps are encapsulated in `scikit-learn Pipeline` objects so that every transformation is fitted exclusively on training-fold data, preventing any data leakage. The full pipeline is: **(1)** impute missing values, **(2)** standardise continuous features (zero mean, unit variance), **(3)** encode categorical variables, **(4)** fit a penalised linear model.

We compared eight model families using 5-fold cross-validated RMSE:

Model	CV RMSE	Notes
ElasticNet (CV-tuned)	23.2	L1 + L2, best balance
LASSO (CV-tuned)	24.1	Pure L1, ultra-sparse
Ridge (CV-tuned)	25.3	L2 only
PCR ($d = 50$ components)	31.8	PCA then OLS
PLS ($d = 15$ components)	32.6	Maximises $\text{Cov}(X\beta, y)$
Gradient Boosting	38.1	High variance at $n = 100$
Random Forest	48.3	Insufficient data for splits
SVR (RBF kernel)	50.1	Kernel trick fails at $n = 100$

Linear penalised models consistently dominate: with $n = p = 100$, tree-based and kernel methods have too few observations to reliably estimate non-linear interactions. PLS and PCR, despite being designed for $p \approx n$, compress X into latent components without zeroing out noise predictors, whereas ElasticNet directly shrinks individual coefficients to zero — a more surgical regularisation when the signal is sparse. ElasticNet minimises

$$\frac{1}{2n} \|y - X\beta\|_2^2 + \alpha \rho \|\beta\|_1 + \frac{\alpha(1 - \rho)}{2} \|\beta\|_2^2,$$

with α and ρ tuned by inner 3-fold CV.

Beyond ElasticNet, we explored two additional sparse methods. **Adaptive LASSO** (Zou 2006) assigns predictor-specific weights $w_j = 1/|\hat{\beta}_j^{\text{Ridge}}|^\gamma$, penalising weak predictors more aggressively than strong ones. Unlike the standard LASSO, it satisfies the *oracle property* (consistent support recovery). With $\gamma = 1.0$ (chosen by grid search over $\{0.25, 0.5, \dots, 3.0\}$), CV RMSE improved to 22.07. **Stability Selection** (Meinshausen & Bühlmann 2010) runs Adaptive LASSO on $B = 100$ sub-samples of size $n/2$, retains only features selected in more than 50% of iterations (≈ 8 stable features), then refits OLS on this support. This reduced the inter-fold standard deviation from 4.69 to 3.92.

Model selection

Hyperparameter selection is performed by nested cross-validation: for each outer fold (5-fold), an inner 3-fold CV loop selects α , ρ , and γ using only the training observations of that fold. The outer fold score is computed on the held-out observations without any further fitting. This strict nesting prevents optimistic bias in the model comparison above.

For the final ensemble, optimal weights $(w_1, w_2) = (0.60, 0.40)$ for Adaptive LASSO and Stability OLS were selected by exhaustive grid search over the out-of-fold predictions of the 5-fold CV run, without re-running any fold.

Missing values

We benchmarked three strategies, all evaluated via 5-fold CV RMSE:

Strategy	CV RMSE	Gain vs. median
Median imputation	33.2	—
KNN imputation ($k = 5$)	26.2	−21%
MICE ($t = 2$)	23.2	−31%

MICE (Multiple Imputation by Chained Equations) was implemented via `IterativeImputer` with a `BayesianRidge` regressor and $t = 2$ iterations. At each iteration, every feature with missing values is regressed on all others and missing entries are replaced by predicted values. Median imputation ignores all inter-feature correlations; KNN captures local structure; MICE models the full joint distribution. The 31% RMSE reduction confirms that imputation quality is the single most impactful modelling decision in this dataset.

Factor handling

Column `C_02` has zero variance (constant value 72) and is dropped before any modelling. The remaining four categorical columns (`C_01`, `C_03–C_05`) take ordered integer values in $\{71, \dots, 75\}$. Although the levels are ordered, we chose *one-hot encoding* rather than ordinal encoding: the spacing between levels is unknown and need not be uniform, so treating the codes as numeric imposes a potentially false metric. With 5 levels per feature, one-hot encoding adds only $4 \times 4 = 16$ binary columns, bringing the total dimensionality to 115 after encoding. Missing categorical values are imputed by mode before encoding. The option `handle_unknown='ignore'` silently zeros out any unseen level in the test set.

Model validation

We used **5-fold cross-validation** (fixed `random_state=42`) throughout. At each fold: **(i)** MICE is fitted on the 80 training observations only; **(ii)** scaling parameters and encoder are fitted on the same 80 observations; **(iii)** hyperparameters are tuned via an inner 3-fold loop on those 80 observations;

(iv) predictions are generated for the 20 held-out observations using the pipeline fitted in steps (i)–(iii). No information from held-out observations enters any fitting step, guaranteeing an unbiased EPE estimate.

We chose 5 folds over 10 because MICE is computationally expensive and each 5-fold held-out set still contains 20 observations, sufficient for a reliable fold-level RMSE. The 5-fold CV estimate carries a slight upward bias (each fold trains on 80 rather than 100 observations), so our estimate is conservative.

The per-fold RMSE of the final ensemble components was:

Fold	Adap. LASSO RMSE	Stab. OLS RMSE
1	27.69	23.36
2	16.25	25.22
3	26.64	26.07
4	18.76	15.71
5	18.77	20.80
Mean $\pm$ std	21.6 $\pm$ 4.7	22.2 $\pm$ 3.9

The two models make complementary errors: Adaptive LASSO excels on Folds 2 and 5, while Stability OLS excels on Folds 1 and 4. Their weighted combination (0.60/0.40) achieves $\widehat{\text{RMSE}} = 21.18$ with $\hat{\sigma} = 3.70$, which is lower than either component alone.

Results

Our final predictor is a **weighted ensemble**:

$$\hat{y} = 0.60 \hat{y}^{\text{AdapLASSO}} + 0.40 \hat{y}^{\text{StabOLS}},$$

where both components use the same MICE pipeline (BayesianRidge, $t = 2$, `random.state=42`). The Adaptive LASSO uses Ridge-initialized weights with $\gamma = 1.0$. The Stability OLS component retains 8 features selected in more than 50% of $B = 200$ bootstrap sub-samples and refits OLS on this support.

Three features are always selected (frequency = 1.00): `x_32` (coefficient 56.7), `x_62` (34.3), `x_36` (27.5). Five additional features appear in 63%–81% of sub-samples. The remaining 107 features receive zero weight, confirming a sparse true signal.

The RMSE progression across methodological steps:

Step	Change	CV RMSE	$\hat{\sigma}$
Baseline	Median + ElasticNet	33.20	5.19
Step 1	KNN imputation ($k = 5$)	26.22	5.07
Step 2	MICE ($t = 2$)	23.20	2.86
Step 3	Adaptive LASSO ($\gamma = 1$)	22.07	4.69
Step 4	Stability Selection + OLS	22.15	3.92
Step 5	Ensemble (0.60/0.40)	21.18	3.70

Applied to the 1,000 test observations, the ensemble produces predictions in $[-14.2, 428.7]$ with mean 199.2 (training mean: 211.5).

Our estimate of the test RMSE is

$$\widehat{\text{RMSE}} = \mathbf{21.18}, \quad \hat{\sigma} = 3.70,$$

corresponding to an approximate 95% interval of $[17.5, 24.9]$. We report the conservative 5-fold CV score rather than a resubstitution error, in keeping with the principle that it is better to be honestly mediocre than confidently wrong.

AI-aid use declaration

We used ChatGPT and Claude as stylistic, grammar, and spelling correctors when writing this report. We also used ChatGPT to clarify technical doubts and solve software-related problems during model design, training, and evaluation.

All results reported were produced by running the code locally on our machines. We reviewed, understood, and take full responsibility for all modelling choices and conclusions presented in the report.